

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Semester :V

Branch : Electronics and Tele-communication Engineering/ Electronics and Communication Engineering

Subject Code & Name: BTETPE504A Analog Circuits

Max Marks: 60

Date:08-01-24

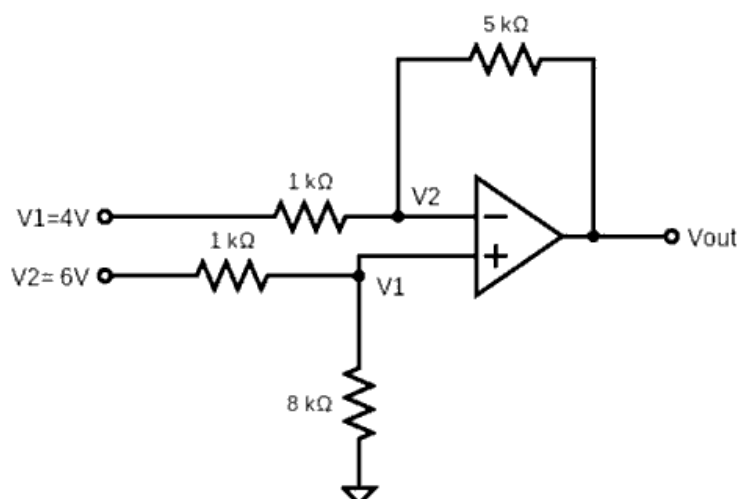
Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	CO	Marks
Q.1 Solve Any Two of the following.		12
A) Give the characteristics of Ideal Op-amp?	CO1	6
B) Explain the different types of feedback in Operational Amplifier?	CO6	6
C) What is an operational amplifier? Explain the Operational Amplifier with the help of its block diagram?	CO3	6

Q.2 Solve Any Two of the following.		12
A) Derive an expression for Non-Inverting amplifier?	CO6	6
B) Calculate the output voltage V_o for the circuit shown	CO3	6



C) Write a short note on Differentiator?	CO6	6
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Q.3 Solve Any Two of the following.		12
A) Explain Positive Precision full Wave Rectifier?	CO2	6
B) Explain the Inverting Comparator?	CO2	6
C) Explain Non-inverting zero crossing detector?	CO2	6

Q.4 Solve Any Two of the following.		12
A) Discuss the factors affecting stability of oscillators?	CO5	6
B) Discuss square wave oscillator?	CO5	6
C) Classify different types of oscillator?	CO8	6
Q.5 Solve Any Two of the following.		12
A) Explain Single Slope Analog to Digital Converter?	CO2	6
B) Write a short note on: Voltage to current converter using op-amp with Ground Load	CO7	6
C) Construct the R-2R ladder network using the following information and find the output voltage of R-2R ladder network where input voltage is 10V and hexadecimal letter "B" is equal to the binary code "1011" in binary is given as digital input. Also find the resolution of the converter?	CO8	6

***** End *****